SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED: CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation

Models available:

- Viola-Jones: 78%, low latency
- YOLO: 92%, moderate latency
- Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

ANS:

Each model must be checked against all three constraints — latency ≤ 100 ms, accuracy ≥ 85%, and moderate hardware compatibility — before it can be considered acceptable.

Model	Accuracy	Latency	Hardware Fit	Constraint Check
Viola-Jones	78%	Low	Fits moderate hardware	Fails accuracy constraint
YOLO	92%	Moderate	Fits moderate hardware	Satisfies all constraints
Deep Learning	95%	High	Needs high-end hardware	Fails latency and hardware constraints

Comparison and Evaluation:

Viola-Jones offers very low latency and is easy to run on moderate hardware, but its accuracy of 78% falls below the 85% requirement, making it unacceptable regardless of its speed advantage.

Deep Learning provides the highest accuracy at 95%, comfortably meeting the accuracy requirement. However, its high latency is likely to exceed the 100 ms limit, causing the system to fail the latency condition. Its heavy computational demand also does not align with hardware that supports only moderate computation.

YOLO meets the accuracy requirement with 92%, well above the 85% threshold. Its moderate latency keeps it within the 100 ms limit, and it is designed to run efficiently on moderate computational hardware, satisfying every stated condition simultaneously.

Decision:

YOLO should be selected for the smart surveillance system because it is the only model that satisfies all three constraints together — acceptable accuracy, latency within the 100 ms limit, and compatibility with moderate-compute hardware.

Conclusion:

Viola-Jones fails on accuracy and Deep Learning fails on latency and hardware suitability, leaving YOLO as the only model that fully complies with the surveillance system's operating conditions. YOLO is therefore the logically justified choice.

2. Autonomous Vehicle Detection Logic

A vehicle detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms

Options:

- Method A: 88% accuracy, 30 ms
- Method B: 93% accuracy, 70 ms
- Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

ANS:

Both conditions — accuracy above 90% and response time under 50 ms — must be satisfied together, since the system prioritizes safety and cannot compromise on either constraint.

Method	Accuracy	Response Time	Accuracy Check (>90%)	Time Check (<50ms)
Method A	88%	30 ms	Fails	Passes
Method B	93%	70 ms	Passes	Fails
Method C	91%	45 ms	Passes	Passes

Comparison and Evaluation:

Method A responds quickly at 30 ms, but its accuracy of 88% is below the 90% safety threshold, making it unsuitable for a system where safety is the top priority.

Method B achieves the highest accuracy at 93%, satisfying the safety requirement. However, its response time of 70 ms exceeds the 50 ms limit, which could delay critical detection decisions in a fast-moving vehicle environment.

Method C balances both requirements effectively, with 91% accuracy exceeding the safety threshold and a 45 ms response time comfortably within the real-time limit.

Decision:

Method C should be selected because it is the only option that satisfies both the accuracy and response-time constraints simultaneously, making it the safest and most practical choice for autonomous vehicle detection.

Conclusion:

Since autonomous driving requires both high accuracy and fast response for safety, Method C offers the best constraint-satisfying trade-off, while Method A and Method B each fail one of the two critical requirements.

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

- Optical Flow: accurate but noisy in dense scenes
- Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

ANS:

The two techniques differ mainly in how they behave as traffic density increases, which directly affects their suitability for a dense-traffic monitoring scenario.

Factor	Optical Flow	Motion Estimation
Motion detail	High	Lower
Accuracy in sparse traffic	High	Moderate
Stability in dense traffic	Low (noisy)	High
Robustness	Degrades under density	Consistent under density

Comparison and Evaluation:

Optical Flow captures fine-grained motion direction and is highly accurate when traffic is light, but as vehicle density increases, overlapping motion vectors introduce noise, reducing its reliability for consistent direction tracking.

Motion Estimation provides less detailed output but maintains stable performance even as the number of moving objects grows, since it relies on broader motion patterns rather than pixel-level vector precision.

Decision:

For high-density traffic conditions, Motion Estimation is the more suitable approach because its stability is not significantly affected by the number of vehicles, ensuring dependable monitoring even during peak congestion.

In lower-density conditions, Optical Flow remains preferable for capturing precise motion direction, so a hybrid design that switches methods based on detected traffic density would offer the best overall performance.

Conclusion:

Motion Estimation is the scenario-appropriate choice for dense traffic because robustness matters more than fine detail when congestion is high, while Optical Flow's noise sensitivity makes it unreliable in exactly the conditions this system must handle.

4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

- System must maintain ≥80% accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

ANS:

The system's accuracy must be checked against the 80% minimum requirement in every scenario, not just on average, since consistent performance across all conditions is required.

Scenario	GMM Accuracy	Requirement (≥80%)	Result
Static scenes	90%	≥80%	Meets requirement
Dynamic scenes	65%	≥80%	Fails requirement

Comparison and Evaluation:

In static scenes, GMM performs well with 90% accuracy, comfortably exceeding the 80% requirement because the background pixel distribution remains stable and predictable over time.

In dynamic scenes, accuracy drops sharply to 65%, well below the required threshold. This decline typically occurs because moving background elements such as foliage, water, or lighting changes are misclassified as foreground, or genuine foreground objects blend into a constantly shifting background model.

Decision:

GMM alone does not meet the requirement, since it fails the 80% threshold in dynamic scenes even though it satisfies it in static scenes. The requirement demands consistency across all scenarios, and a single failing case is enough to disqualify GMM as a standalone solution.

A practical improvement would be to combine GMM with complementary techniques such as adaptive learning rates, morphological filtering, or a deep-learning-based detector to stabilize performance specifically in dynamic scenes.

Conclusion:

GMM alone is insufficient because its dynamic-scene accuracy of 65% falls short of the 80% requirement. An enhanced or hybrid background modeling approach is necessary to achieve consistent accuracy across both static and dynamic conditions.

5. Depth Estimation Strategy

Constraints:

- Only one camera available
- Moderate depth accuracy acceptable

Options:

- Stereo Vision: high accuracy, needs two cameras
- Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

ANS:

Feasibility here depends first on hardware compatibility with the single-camera constraint, and only then on how well the accuracy level matches what is acceptable.

Factor	Stereo Vision	Structure from Motion
Camera requirement	Two cameras	Single camera
Accuracy	High	Moderate
Hardware constraint fit	Fails (needs 2 cameras)	Passes (single camera)
Accuracy requirement fit	Exceeds requirement	Meets requirement

Comparison and Evaluation:

Stereo Vision delivers the highest depth accuracy by using two cameras to triangulate distance, but this directly conflicts with the constraint of only one available camera, making it hardware-infeasible regardless of its accuracy advantage.

Structure from Motion estimates depth using a sequence of frames from a single moving camera, which fits the available hardware exactly. Its accuracy is moderate rather than high, but the requirement explicitly states that moderate depth accuracy is acceptable.

Decision:

Structure from Motion should be selected because it is the only approach that is feasible under the single-camera hardware constraint, and its moderate accuracy is sufficient to meet the stated requirement.

Conclusion:

Since Stereo Vision cannot be deployed without a second camera, Structure from Motion is the logically feasible choice — it satisfies the hardware limitation while still providing an acceptable level of depth accuracy.

Code of Conduct:

I G.Joel Richard (192524054) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ______________________

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation